1.1 Are X and Y as defined in the following table independently distributed, why?

x	a	a	b	b
y	a	b	a	b
$p(X = x, Y = y)$	0.32	0.08	0.48	0.12

1.2 Compute the entropies $H(X), H(Y), H(X|Y), H(Y|X), H(X, Y)$. Use the following log_2 values:

k	0.08	0.12	0.2	0.32	0.4	0.48	0.6	0.8
$log_2(k)$	-3.644	-3.059	-2.322	-1.644	-1.322	-1.059	-0.737	-0.322

1.3 Compute the Kullback-Leibler divergence $D(X||Y)$.

Hints: $H(X) = -\sum_{x \in X} p(x) \lg p(x)$ $\quad H(Y|X) = -\sum_{x \in X} \sum_{y \in Y} p(x, y) \lg p(y|x)$ $\quad H(X, Y) = -\sum_{x \in X} \sum_{y \in Y} p(x, y) \lg p(x, y)$

5+(5*2)+5=20

2 A pattern C $\{yes, no\}$ to recognize a name event N $\{name, not_name\}$ has the following properties:

$$P(C = yes|N = name) = 0.9$$
$$P(C = yes|N = not_name) = 0.2$$

Assume the following:

- In newspaper text, around 5% of the words are names.
- In scientific text, around 1% of the words are names.

a) What is the probability to really see a name if C says so?

b) How low must the false positive rate P(C=yes|N=not-name) get so that this probability goes up to 50% for both kinds of text?

5+5=10